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K J Somaiya School of Engineering
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Module 3 Unsupervised Learning – (PCA)

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What is the curse of dimensionality?

Try building several machine learning models to analyze the performance of a Formula One (F1) driver.

Consider the following cases:

Model_1 - 2 features (circuit name, country name)

Model_2 - 4 features (circuit name, country name, weather, max speed of the car)

Model_3 - 8 features (circuit name, country name, weather, max speed of the car , say driver's experience, number of wins, car condition, driver's physical fitness)

Model_4 - 16 features (circuit name, country name, weather, max speed of the car , say driver's experience, number of wins, car condition, driver's physical fitness, age, latitude, longitude, driver's height, hair color, car color, the car company, driver's marital status)

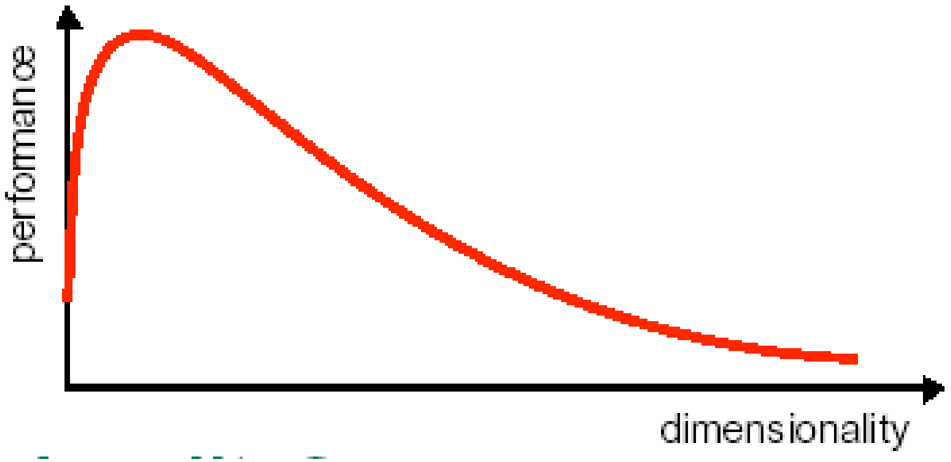
Model_5 - 32 features, **Model_6** - 64 features, **Model_7** - 128 features, **Model_8** - 256 features, **Model_9** - 512 features, **Model_10** - 1024 features.

What is the curse of dimensionality?

- Assuming the training data remains constant, it is observed that on increasing the number of features the accuracy tends to increase until a certain threshold value and after that, it starts to decrease.
- The accuracy of Model_1 < accuracy of Model_2 < accuracy of Model_3 but if we try to extrapolate this trend it doesn't hold true for all the models having more than 8 features.
- Now you might wonder if we are providing some extra information for the model to learn why is it so that the performance starts to degrade.

What is the curse of dimensionality?

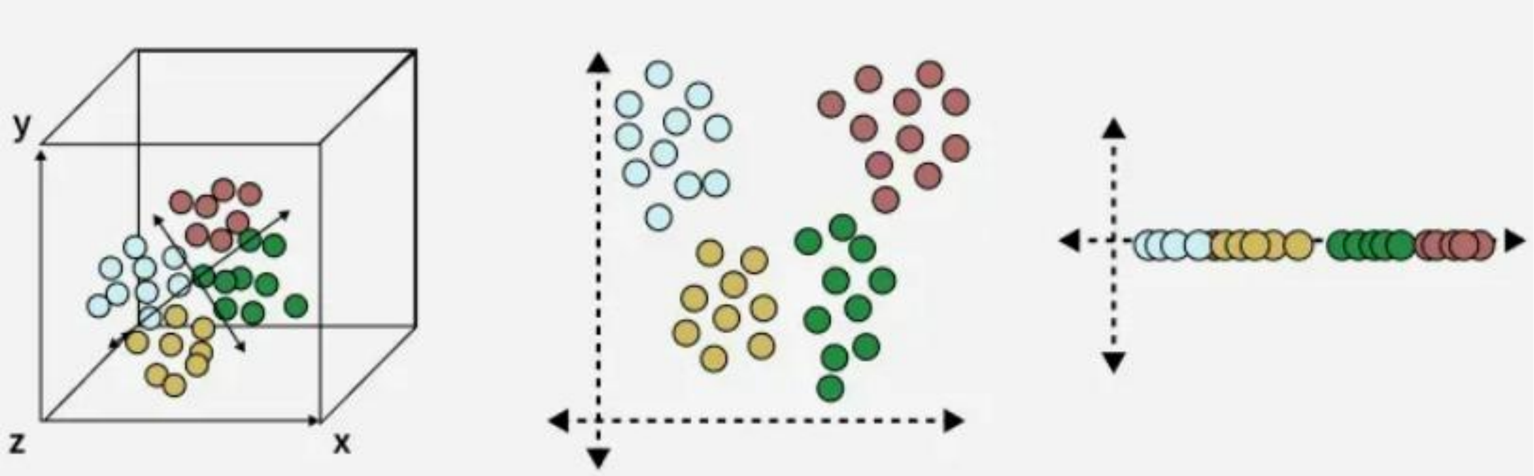
- For a given sample size, there is a maximum number of features above which the performance of classifier will degrade rather than improve.



- The curse of dimensionality was first termed by Richard E. Bellman when considering problems in dynamic programming.

What is Dimensionality Reduction

- Dimensionality Reduction is the process of reducing the number of input variables (features) in a dataset while preserving as much important information as possible.



Why Dimensionality Reduction?

- Prevent – Curse of Dimensionality
- Improve the performance of the model
- Visualize the data – understand the data

Dimensionality Reduction

Advantages

- Simplify the pattern representation and the classifiers
- Faster classifier with less memory consumption
- Alleviate curse of dimensionality with limited data sample

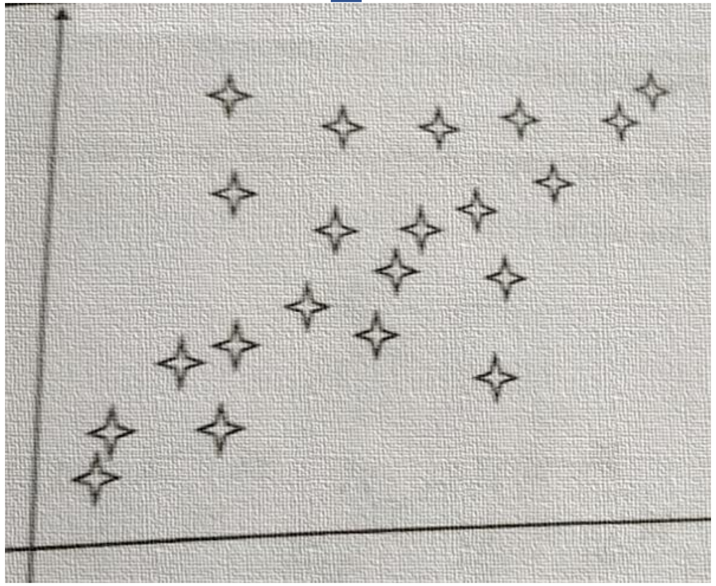
Disadvantages

- Loss information
- Increased error in the resulting recognition system

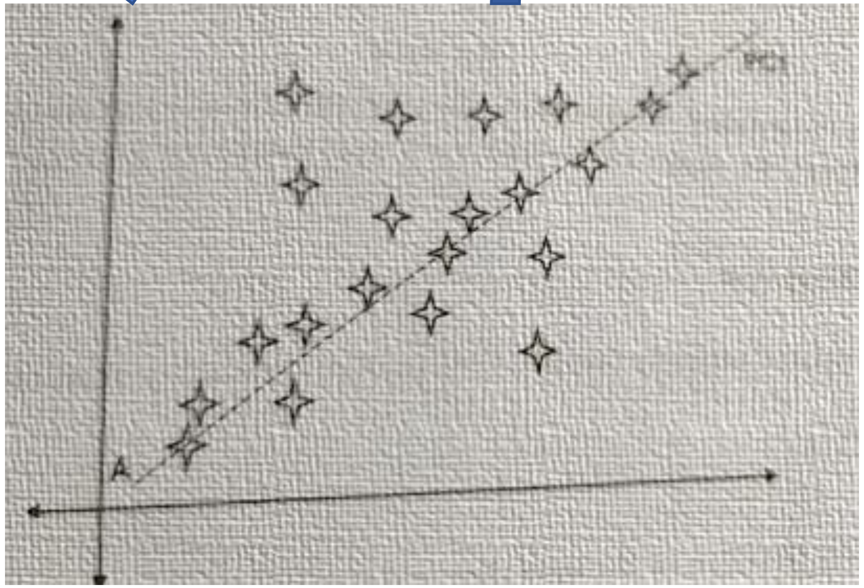
Feature Reduction: PCA

- PCA (Principal Component Analysis) is an unsupervised dimension reduction method.
- It aims at finding the most important dimensions (called principal components) for the given dataset.
- A principal component is a direction in which the data has the largest variation.

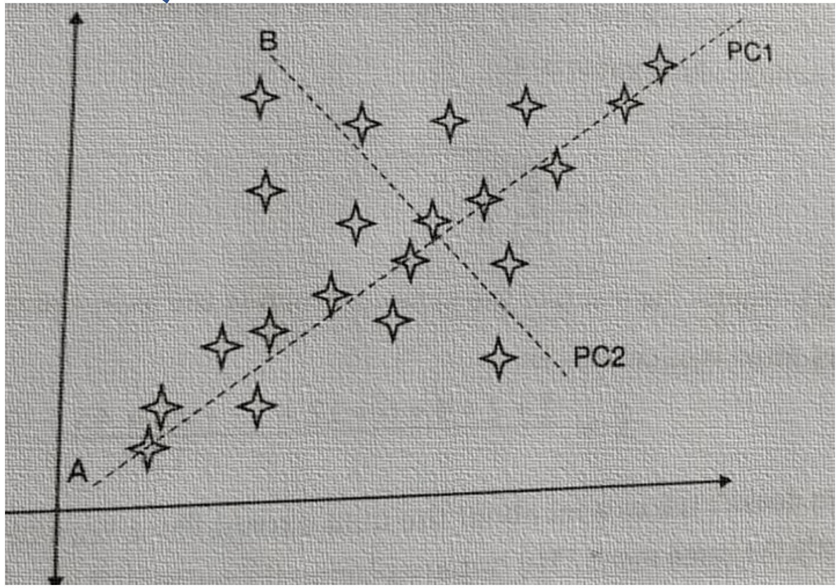
PCA Conceputal Steps:



Centre the data by subtracting off the mean



Choose the direction with largest variation, place an axis in that direction



Look at the remaining variations, find another axis that is that is orthogonal to the first axis and covers maximum variations

Statistical Concepts Required for PCA

1. Mean – the mean of a dataset is:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Where

xi = data points

n = number of samples

Statistical Concepts Required for PCA

2. Standard deviation measures the spread of data around the mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{n-1}} \qquad s = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{(n-1)}}$$

Large value → data widely spread
Small value → data tightly clustered.

Statistical Concepts Required for PCA

3. Variance is square of the standard deviation.

$$Var(X) = \sigma^2$$

$$var(X) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})}{(n-1)}$$

It measure how far data points are from the mean.

Statistical Concepts Required for PCA

- Covariance measures how two variables vary together.

$$Cov(X,Y) = \frac{\sum(x_i - \mu_x)(y_i - \mu_y)}{n-1}$$

Covariance Value	Meaning
Positive	Variables increase together
Negative	One increases while other decreases
Zero	No relationship

Covariance Matrix

Representing Covariance between dimensions as a matrix e.g. for 3 dimensions:

$$C = \begin{bmatrix} cov(x,x) & cov(x,y) & cov(x,z) \\ cov(y,x) & cov(y,y) & cov(y,z) \\ cov(z,x) & cov(z,y) & cov(z,z) \end{bmatrix}$$

- Diagonal is the variances of x, y and z
- cov(x,y) = cov(y,x) hence matrix is symmetrical about the diagonal
- N-dimensional data will result in NxN covariance matrix

Covariance

- Exact value is not as important as it’s sign.
- (+) A positive value of covariance indicates both dimensions increase or decrease together e.g. as the number of hours studied increases, the marks in that subject increase.
- (-) A negative value indicates while one increases the other decreases, or vice-versa e.g. time spent on social media vs performance in exam.
- (zero) If covariance is zero: the two dimensions are independent of each other e.g. heights of students vs the marks obtained in a subject

Eigen vectors

- $M = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 5 \end{bmatrix}$ The resulting vector is not integer multiple of the original vector
- $N = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4 \times \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ The resulting vector is exactly 4 times the original vector. This is called eigenvector.

- *We can transform and change matrices into new vectors by multiplying a matrix with a vector. The multiplication of the matrix by a vector computes a new vector. This is the transformed vector.*

Eigenvector

- If the new transformed vector is just a scaled form of the original vector then the original vector is known to be an **eigenvector** of the original matrix
- An eigenvector is a vector that does not change when a transformation is applied to it, except that it becomes a scaled version of the original vector.
- **Eigenvalue**—The scalar that is used to transform (stretch) an Eigenvector.

Eigenvector and Eigenvalue

- An eigenvector **does not change direction** in a transformation:
- In this shear mapping the red arrow changes direction, but the blue arrow does not. The blue arrow is an eigenvector of this shear mapping because it does not change direction, and since its length is unchanged, its eigenvalue is 1.
- **1** means no change,
- **2** means doubling in length,
- **−1** means pointing backwards along the eigenvalue's direction

The Mathematics

- For a square matrix **A**, an Eigenvector and Eigenvalue make this equation true:

$$\underbrace{A}_{\text{Matrix}}\underbrace{v}_{\text{Eigenvector}} = \underbrace{\lambda}_{\text{Eigenvalue}}\underbrace{v}_{\text{Eigenvector}}$$

Av gives us:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -6\times1+3\times4 \\ 4\times1+5\times4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

λv gives us :

$$6\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

Yes they are equal! So $Av = \lambda v$ as promised.

How do we find these eigen things?

We start by finding the **eigenvalue**:

- We know this equation must be true: $Av = \lambda v$
- Now let us put in an identity matrix so we are dealing with matrix-vs-matrix:

$$Av = \lambda I v$$

- Bring all to left hand side:

$$Av - \lambda I v = 0$$

- If **v** is non-zero then we can solve for λ using just the determinant:

$$| A - \lambda I | = 0$$

Example for λ

- Start with | A − λI | = 0
- Given matrix

$$A = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}$$

- Substitute into equation

$$\left| \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} -6-\lambda & 3 \\ 4 & 5-\lambda \end{bmatrix} \right| = 0$$

- Calculating the determinant

$$(-6-\lambda)(5-\lambda) - 12 = 0$$

$$(-30 + 6\lambda - 5\lambda + \lambda^2) - 12 = 0$$

$$\lambda^2 + \lambda - 42 = 0$$

- **Final Eigenvalues**

$$\lambda^2 + \lambda - 42 = 0$$

λ1=−7

λ2=6

Find the Eigenvector for Eigenvalue λ=6

- Substitute eigenvalue into the equation:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix}$$

- Matrix multiplication gives

$$-6x + 3y = 6x$$

$$4x + 5y = 6y$$

$$-6x + 3y - 6x = 0$$

$$-12x + 3y = 0$$

$$4x + 5y - 6y = 0$$

$$4x - y = 0$$

- Solve the it

$$4x - y = 0$$

$$y = 4x$$

- Thus the eigenvector is any non-zero multiple of:

$$v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Verify Av=λv

- Compute AV

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} (-6)(1) + (3)(4) \\ (4)(1) + (5)(4) \end{bmatrix}$$

$$= \begin{bmatrix} -6 + 12 \\ 4 + 20 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

- Compute λv

$$6 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

PCA

Principal components analysis (PCA) is a technique that can be used to simplify a dataset

- It is a linear transformation that chooses a new coordinate system for the data set such that greatest variance by any projection of the data set comes to lie on the first axis (then called the first principal component),the second greatest variance on the second axis, and so on.
- PCA can be used for reducing dimensionality by eliminating the later principal components.

Steps Involved in the PCA

Step 1: Standardize the dataset.

Step 2: Calculate the covariance matrix for the features in the dataset.

Step 3: Calculate the eigenvalues and eigenvectors for the covariance matrix.

Step 4: Sort eigenvalues and their corresponding eigenvectors.

Step 5: Pick k eigenvalues and form a matrix of eigenvectors.

Step 6: Transform the original matrix.

Example 1

- Assume we have the below dataset which has 2 features and a total of 4 training examples.

Size (X_1)	Rooms (X_2)
5	10
3	7
10	15
2	4

Example 1

1. Standardize the Dataset

- First, we need to standardize the dataset and for that, we need to calculate the mean and standard deviation for each feature.

Size (X_1)	Rooms (X_2)
5	10
3	7
10	15
2	4

$$\bar{X}_1 = \frac{5+3+10+2}{4} = 5$$

$$\bar{X}_2 = \frac{10+7+15+4}{4} = 9$$

$$\sigma_1 = \sqrt{\frac{(5-5)^2+(3-5)^2+(10-5)^2+(2-5)^2}{4}} = \sqrt{\frac{0+4+25+9}{4}} = \sqrt{9.5} = 3.082$$

$$\sigma_2 = \sqrt{\frac{(10-9)^2+(7-9)^2+(15-9)^2+(4-9)^2}{4}} = \sqrt{\frac{1+4+36+25}{4}} = \sqrt{16.5} = 4.062$$

Example 1

- New data

$$x_{new} = \frac{x - \mu}{\sigma}$$

Size (X_1)	Rooms (X_2)
5	10
3	7
10	15
2	4

Obs	(Z1 = (X1-5)/3.082)	(Z_2 = (X2-9)/4.062)
1	0	0.246
2	-0.649	-0.492
3	1.622	1.478
4	-0.973	-1.231

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Example 1

Obs	(Z1 = (X1-5)/3.082)	(Z_2 = (X2-9)/4.062)
1	0	0.246
2	-0.649	-0.492
3	1.622	1.478
4	-0.973	-1.231

Step 2. Calculate the covariance matrix for the whole dataset

The formula to calculate the covariance matrix:

	Z1	Z2
Z1	Var(Z1)	COV(Z1,Z2)
Z2	COV(Z2,Z1)	Var(Z2)

- var(z1) = ((0-0)² + (-0.649-0)² + (1.622-0)² +(0.973–0)²)/4=3.99/4

var (z1) = 1

- cov(z1,z2) =((0)(0.246)+(−0.649)(−0.492)+(1.622)(1.478)+(−0.973)(−1.231))/3

=3.914/3

$$C = \begin{bmatrix} 1 & 1.305 \\ 1.305 & 1 \end{bmatrix}$$

- **cov(z1,z2) = 1.305**

Example 1

3. Calculate eigenvalues and eigen vectors.

$Av-\lambda Iv = 0$;

$(A-\lambda I)v = 0$

Since we have already know v is a non- zero vector, only way this equation can be equal to zero, if $\det(A-\lambda I) = 0$

Example 1

- $(A-\lambda I)v = 0$

$$\begin{vmatrix} 1-\lambda & 1.305 \\ 1.305 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - (1.305)^2 = 0$$

$$(1-\lambda)^2 - 1.703 = 0$$

$$(1-\lambda) = \pm\sqrt{1.703} = \pm 1.305$$

- Eigen values

$$\lambda_1 = 1 + 1.305 = 2.305$$

$$\lambda_2 = 1 - 1.305 = -0.305$$

Example 1

- For λ1=2.305

$$\begin{bmatrix} -1.305 & 1.305 \\ 1.305 & -1.305 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- For λ2=−0.305

$$\begin{bmatrix} 1.305 & 1.305 \\ 1.305 & 1.305 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example 1

$$\begin{bmatrix} -1.305 & 1.305 \\ 1.305 & -1.305 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1.305 & 1.305 \\ 1.305 & 1.305 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

- A **unit vector** is a vector whose **magnitude (length) is exactly 1**.

For a vector $v = [v_1, v_2, \ldots, v_n]$:

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

If:

$$\|v\| = 1$$

then v is a **unit vector**.

For $v_1 = [1, 1]$:

$$\|v_1\| = \sqrt{2} \neq 1$$

- For normalized

$$v_1^{norm} = \frac{1}{\sqrt{2}}[1, 1]$$

Example 1

- **Step 4: Sort Eigenvalues**

Eigenvalue	Eigenvector
2.305	(v1)
-0.305	(v2)

- **Step 5: Select Top k Components**
- Choose k=1(maximum variance)

V1 = [$1/\sqrt{2}$, $1/\sqrt{2}$]

Example 1

Step 6 Transform the original matrix.

Feature matrix * top k eigenvectors = Transformed Data

Obs	(Z1 = (X1-5)/3.082)	(Z2 = (X2-9)/4.062)
1	0	0.246
2	-0.649	-0.492
3	1.622	1.478
4	-0.973	-1.231

$$* \quad \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \quad =$$

PC1
0.174
-0.806
2.192
-1.558

Example 1 Python

Steps Involved in the PCA

Step 1: Standardize the dataset.

Step 2: Calculate the covariance matrix for the features in the dataset.

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Step 4: Sort eigenvalues and their corresponding eigenvectors.

Step 5: Pick k eigenvalues and form a matrix of eigenvectors.

Step 6: Transform the original matrix.

Example 2

- Let’s Say we have marks of 6 students in 3 subjects.

Student	Math	Science	English
A	85	80	78
B	90	85	88
C	78	70	72
D	92	88	85
E	70	65	68
F	88	82	80

Example 2 –Standardize the Data

$$Z = \frac{X - \text{mean}(X)}{\text{std}(X)}$$

- Original

Student	Math	Science	English
A	85	80	78
B	90	85	88
C	78	70	72
D	92	88	85
E	70	65	68
F	88	82	80
Mean	83.83	78.33	78.5
Std	7.62	8.18	6.92

New

Student	Math	Science	English
A	0.153008	0.203785	-0.072232
B	0.808755	0.815139	1.372399
C	-0.765039	-1.018924	-0.939010
D	1.071054	1.181952	0.939010
E	-1.814235	-1.630278	-1.516862
F	0.546456	0.448327	0.216695

Example 2 – Covariance Matrix

	Math	Science	English
Math	VAR(M)	COV(M,S)	COV(M,E)
Science	COV(S,M)	VAR(S)	COV(S,E)
English	COV(E,M)	COV(E,S)	VAR(E)

- Since you have already standardized the features, you can consider Mean = 0 and Standard Deviation=1 for each feature.

Student	Math	Science	English
A	0.153008	0.203785	-0.072232
B	0.808755	0.815139	1.372399
C	-0.765039	-1.018924	-0.939010
D	1.071054	1.181952	0.939010
E	-1.814235	-1.630278	-1.516862
F	0.546456	0.448327	0.216695

Var(M)

$= ((0.153008-0)^2+(0.808755-0)^2+(-0.76504-0)^2+(1.071054-0)^2+(-1.81424-0)^2 + (0.546456-0)^2)/6$

$=6.0/5=1.2$

COV(M,S)

$= ((0.153008-0)+(0.808755-0)^2+(-0.76504-0)^2+(1.071054-0)^2+(-1.81424-0)^2 + (0.546456-0)^2)/6$

$=5.938578 /5=1.187716$

$$\text{Covariance}=\begin{vmatrix} 1.2 & 1.187716 & 1.1387 \\ 1.187716 & 1.2 & 1.1481 \\ 1.13867 & 1.148135 & 1.2 \end{vmatrix}$$

Example 2 – Eigenvalues and Eigen Vectors

Calculate eigenvalues and eigen vectors.

$Av - \lambda I v = 0$;

$(A - \lambda I)v = 0$

Since we have already know v is a non- zero vector, only way this equation can be equal to zero, if $\det(A - \lambda I) = 0$

Example 2 Eigenvalues

1.2- λ	1.187716	1.1387
1.187716	1.2- λ	1.1481
1.13867	1.148135	1.2- λ

- When you solve the following the matrix by considering 0 on right-hand side, you can define eigen values as

$$\boldsymbol{\lambda} = 3.51647796\ ,0.07174828\ ,0.0117737$$

Example 2 – Eigen Vectors

- Then, substitute each eigen value in (A-λI)v=0 equation and solve the same for different eigen vectors

$$\begin{vmatrix} 1.2-\lambda & 1.187716 & 1.1387 \\ 1.187716 & 1.2-\lambda & 1.1481 \\ 1.13867 & 1.148135 & 1.2-\lambda \end{vmatrix} \quad X \quad \begin{vmatrix} V1 \\ V2 \\ V3 \end{vmatrix} \quad = 0$$

- For λ = 3.51647796, solving the above equation using Cramer's rule, the values for the v vector are

v1= -0.5790409, v2= -0.58058703, v3= -0.57239002

Example 2

- We can from a matrix using the **eigen vectors.**

e1	e2	e3
-0.57904	-0.66755	-0.46807
-0.58059	0.740678	-0.3381
-0.57239	-0.07598	0.816454

- Arrange them in descending order and pick up the topmost Eigen values.

Example 2 - Pick k eigenvalues and form a matrix of eigenvectors and Transform the original matrix.

Feature matrix * top k eigenvectors = Transformed Data

Student	Math	Science	English
A	0.153008	0.203785	-0.072232
B	0.808755	0.815139	1.372399
C	-0.765039	-1.018924	-0.939010
D	1.071054	1.181952	0.939010
E	-1.814235	-1.630278	-1.516862
F	0.546456	0.448327	0.216695

6x3

x

e1	e2
-0.57904	-0.66755
-0.58059	0.740678
-0.57239	-0.07598

3x2

=

PC1	PC2
-0.1656	-0.1995
-1.7271	0.46635
1.57204	-0.0641
-1.8439	-0.1343
2.86527	0.16194
-0.7007	-0.2304

6x2

Example 3

1. Standardize the Dataset

- Assume we have the below dataset which has 4 features and a total of 5 training examples.

f1	f2	f3	f4
1	2	3	4
5	5	6	7
1	4	2	3
5	3	2	1
8	1	2	2

Example 3

- First, we need to standardize the dataset and for that, we need to calculate the mean and standard deviation for each feature.

f1	f2	f3	f4
1	2	3	4
5	5	6	7
1	4	2	3
5	3	2	1
8	1	2	2

	f1	f2	f3	f4
μ =	4	3	3	3.4
σ =	3	1.58114	1.73205	2.30217

$$x_{new} = \frac{x - \mu}{\sigma}$$

After applying the formula for each feature in the dataset is transformed as below:

f1	f2	f3	f4
-1	-0.63246	0	0.26062
0.33333	1.26491	1.73205	1.56374
-1	0.63246	-0.57735	-0.17375
0.33333	0	-0.57735	-1.04249
1.33333	-1.26491	-0.57735	-0.60812

Example 3

2. Calculate the covariance matrix for the whole dataset

The formula to calculate the covariance matrix:

	f1	f2	f3	f4
f1	var(f1)	cov(f1,f2)	cov(f1,f3)	cov(f1,f4)
f2	cov(f2,f1)	var(f2)	cov(f2,f3)	cov(f2,f4)
f3	cov(f3,f1)	cov(f3,f2)	var(f3)	cov(f3,f4)
f4	cov(f4,f1)	cov(f4,f2)	cov(f4,f3)	var(f4)

For Population

$$\mathrm{Cov}(x,y) = \frac{\Sigma\,(x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$\mathrm{Cov}(x,y) = \frac{\Sigma\,(x_i - \bar{x}) * (y_i - \bar{y})}{(N-1)}$$

*Since we have standardized the dataset, so the **mean for each feature is 0** and the standard deviation is 1.*

Example 3

f1	f2	f3	f4
-1	-0.63246	0	0.26062
0.33333	1.26491	1.73205	1.56374
-1	0.63246	-0.57735	-0.17375
0.33333	0	-0.57735	-1.04249
1.33333	-1.26491	-0.57735	-0.60812

- var(f1) = $((-1.0-0)^2 + (0.33-0)^2 + (-1.0-0)^2 + (0.33–0)^2 + (1.33–0)^2)/5$
- **var (f1) = 0.8**
- cov(f1,f2) =((-1.0–0)*(-0.632456-0) +
  (0.33–0)*(1.264911-0) +
  (-1.0–0)* (0.632456-0)+
  (0.33–0)*(0.000000 -0)+
  (1.33–0)*(-1.264911–0))/5
- **cov(f1,f2) = -0.25298**

Example3

- In the similar way be can calculate the other covariance's and which will result in the below covariance matrix

	f1	f2	f3	f4
f1	0.8	-0.25298	0.03849	-0.14479
f2	-0.25298	0.8	0.51121	0.4945
f3	0.03849	0.51121	0.8	0.75236
f4	-0.14479	0.4945	0.75236	0.8

Example 3

3. Calculate eigenvalues and eigen vectors.

$Av - \lambda Iv = 0$;

$(A - \lambda I)v = 0$

Since we have already know v is a non- zero vector, only way this equation can be equal to zero, if $\det(A - \lambda I) = 0$

Example 3

3. Calculate eigenvalues and eigen vectors.

$$\begin{bmatrix} 0.8 & -0.25298 & 0.03849 & -0.14479 \\ -0.25298 & 0.8 & 0.51121 & 0.4945 \\ 0.03849 & 0.51121 & 0.8 & 0.75236 \\ -0.14479 & 0.4945 & 0.75236 & 0.8 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = 0$$

	f1	f2	f3	f4
f1	0.8 - λ	-0.25298	0.03849	-0.14479
f2	-0.25298	0.8- λ	0.51121	0.4945
f3	0.03849	0.51121	0.8 - λ	0.75236
f4	-0.14479	0.4945	0.75236	0.8 - λ

Solving the above equation = 0

- ***λ = 2.51579324 , 1.0652885 , 0.39388704 , 0.02503121***

Example 3

- **Eigenvectors:**

Solving the (A-λI)v = 0 equation for v vector with different λ values:

$$\begin{pmatrix} 0.800000-\lambda & -(0.252982) & 0.038490 & -(0.144791) \\ -(0.252982) & 0.800000-\lambda & 0.511208 & 0.494498 \\ 0.038490 & 0.511208 & 0.800000-\lambda & 0.752355 \\ -(0.144791) & 0.494498 & 0.752355 & 0.800000-\lambda \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = 0$$

For λ = *2.51579324, solving the above equation using Cramer's rule, the values for v vector are*

v1 = 0.16195986

v2 = -0.52404813

v3 = -0.58589647

v4 = -0.59654663

Example 3

- Going by the same approach, we can calculate the eigen vectors for the other eigen values. We can from a matrix using the **eigen vectors.**

e1	e2	e3	e4
0.161960	-0.917059	-0.307071	0.196162
-0.524048	0.206922	-0.817319	0.120610
-0.585896	-0.320539	0.188250	-0.720099
-0.596547	-0.115935	0.449733	0.654547

4. **Sort eigenvalues and their corresponding eigenvectors.**

- Since eigenvalues are already sorted in this case so no need to sort them again.

Example 3

5. Pick k eigenvalues and form a matrix of eigenvectors

- If we choose the top 2 eigenvectors, the matrix will look like this:

e1	e2
0.161960	-0.917059
-0.524048	0.206922
-0.585896	-0.320539
-0.596547	-0.115935

6. Transform the original matrix.

Feature matrix * top k eigenvectors = Transformed Data

f1	f2	f3	f4
-1.000000	-0.632456	0.000000	0.260623
0.333333	1.264911	1.732051	1.563740
-1.000000	0.632456	-0.577350	-0.173749
0.333333	0.000000	-0.577350	-1.042493
1.333333	-1.264911	-0.577350	-0.608121

(5,4)

*

e1	e2
0.161960	-0.917059
-0.524048	0.206922
-0.585896	-0.320539
-0.596547	-0.115935

(4,2)

=

nf1	nf2
0.014003	0.755975
-2.556534	-0.780432
-0.051480	1.253135
1.014150	0.000239
1.579861	-1.228917

(5,2)

```
import numpy as np
import pandas as pd
A = np.matrix([[1,2,3,4],
               [5,5,6,7],
               [1,4,2,3],
               [5,3,2,1],
               [8,1,2,2]])

df = pd.DataFrame(A,columns  = ['f1','f2','f3','f4'])
df_std  = (df - df.mean()) / (df.std())
n_components = 2
from sklearn.decomposition import PCA
pca = PCA(n_components=n_components)
principalComponents = pca.fit_transform(df_std)
principalDf = pd.DataFrame(data=principalComponents,columns=['nf'+str(i+1) for i in range(n_components)])
print(principalDf)
```

```
        nf1       nf2
0 -0.014003  0.755975
1  2.556534 -0.780432
2  0.051480  1.253135
3 -1.014150  0.000239
4 -1.579861 -1.228917
```

Solve : Apply feature reduction using PCA

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

Reference

- https://www.simplilearn.com/tutorials/machine-learning-tutorial/principal-component-analysis
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